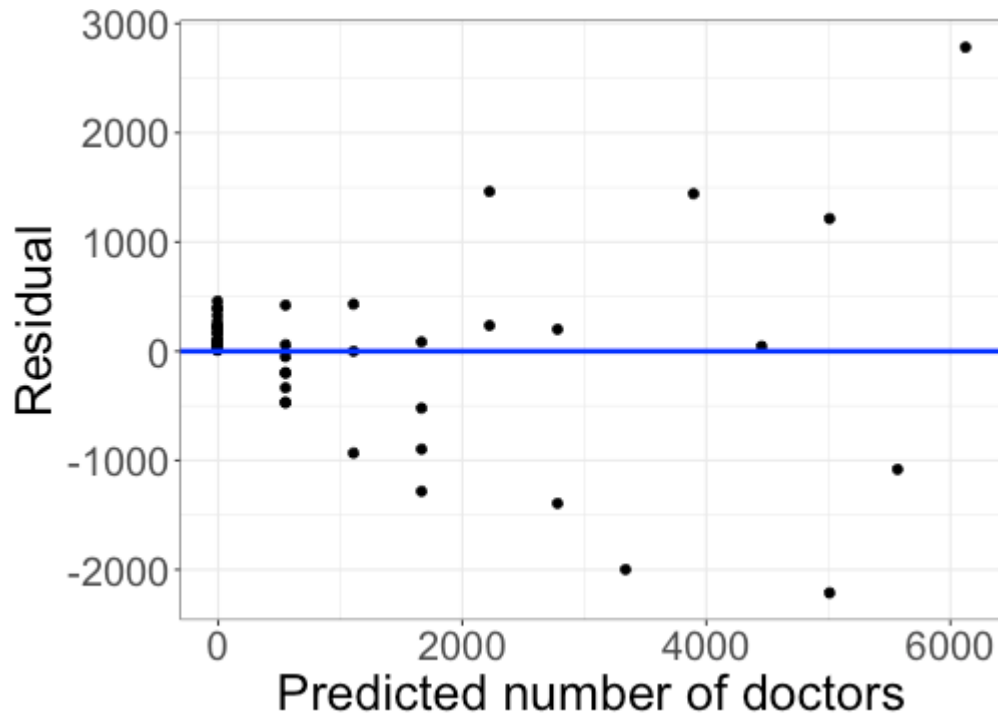


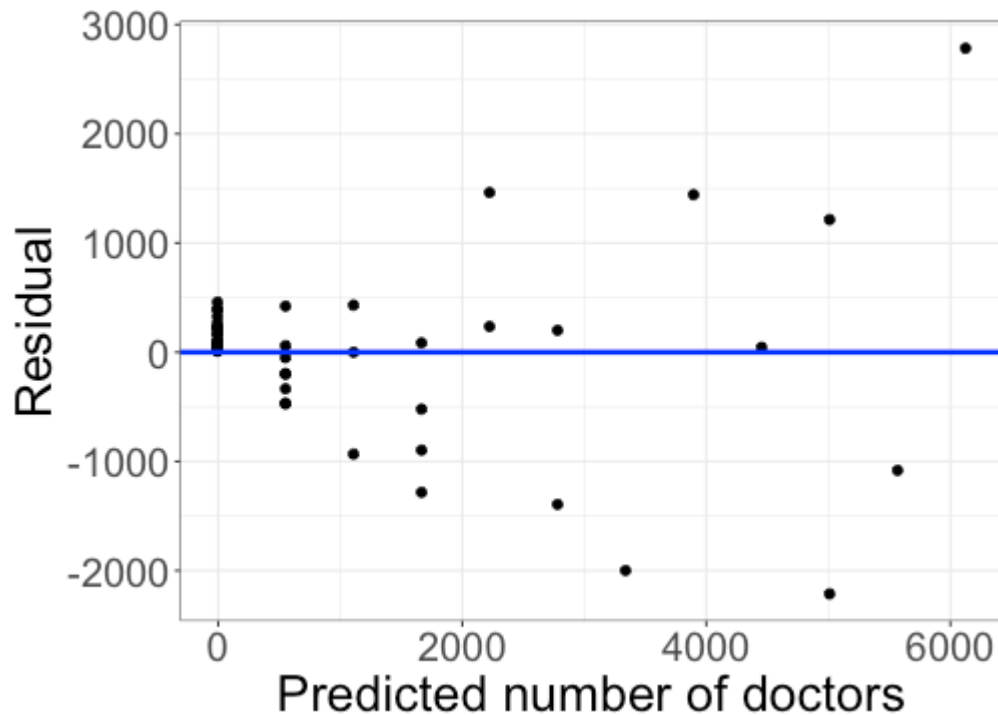
Nonlinear relationships and transformations

Last time



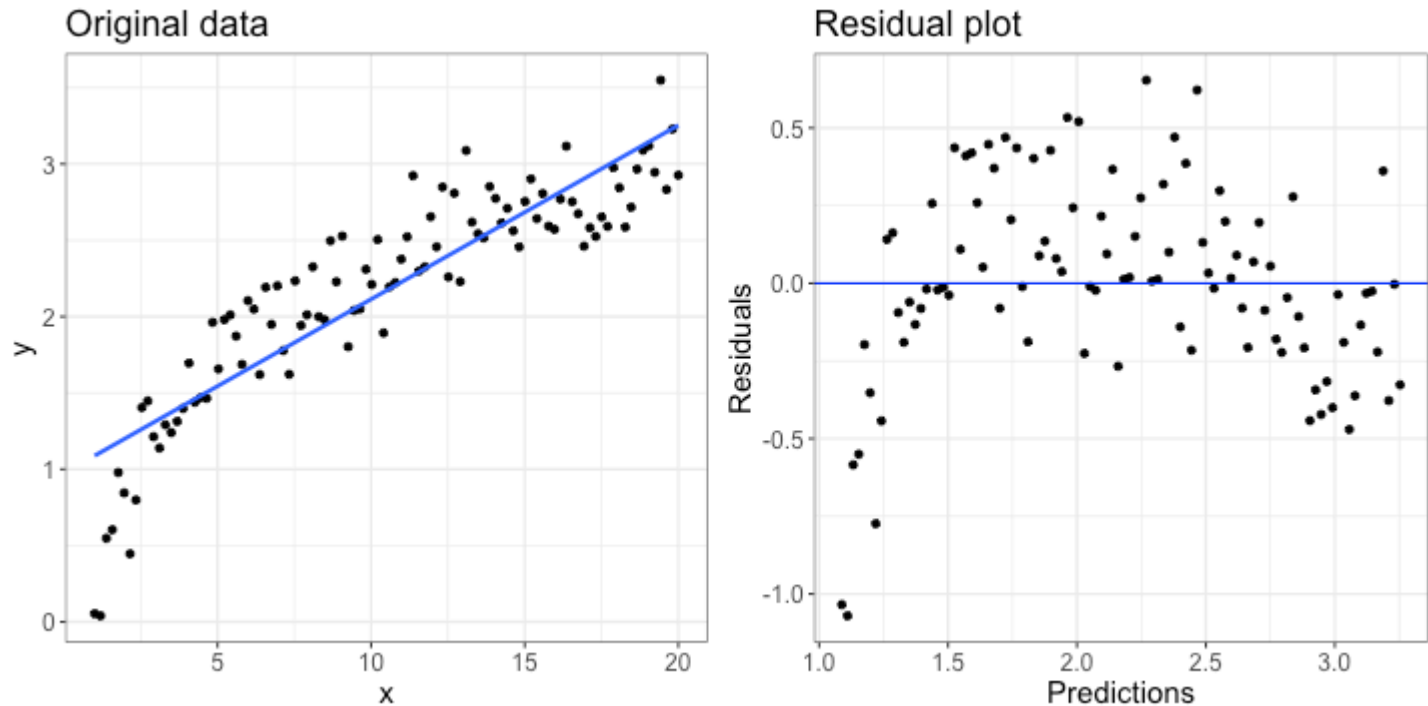
Do the shape and constant variance assumptions look reasonable?

Today



Today: handling non-linear relationships

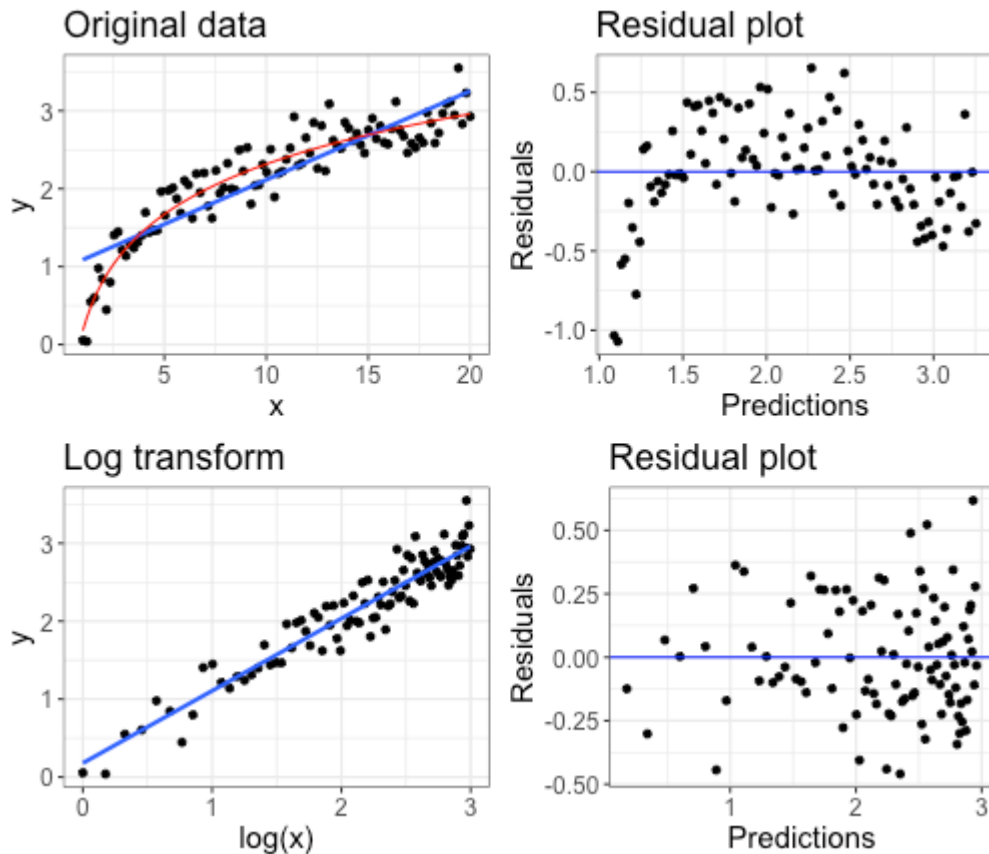
Example



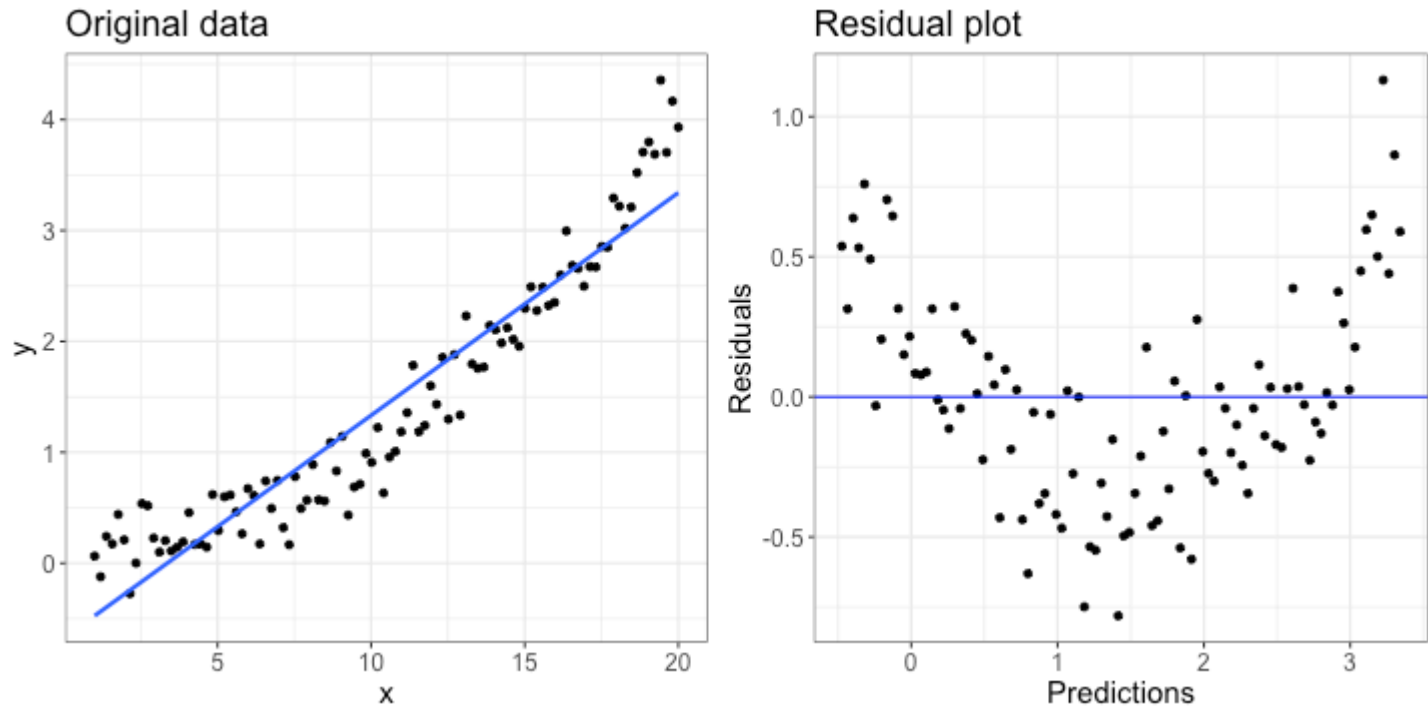
The relationship between x and y does not appear linear. How could we model that relationship instead?

Example: log transform

$$Y = \beta_0 + \beta_1 \log(X) + \varepsilon$$



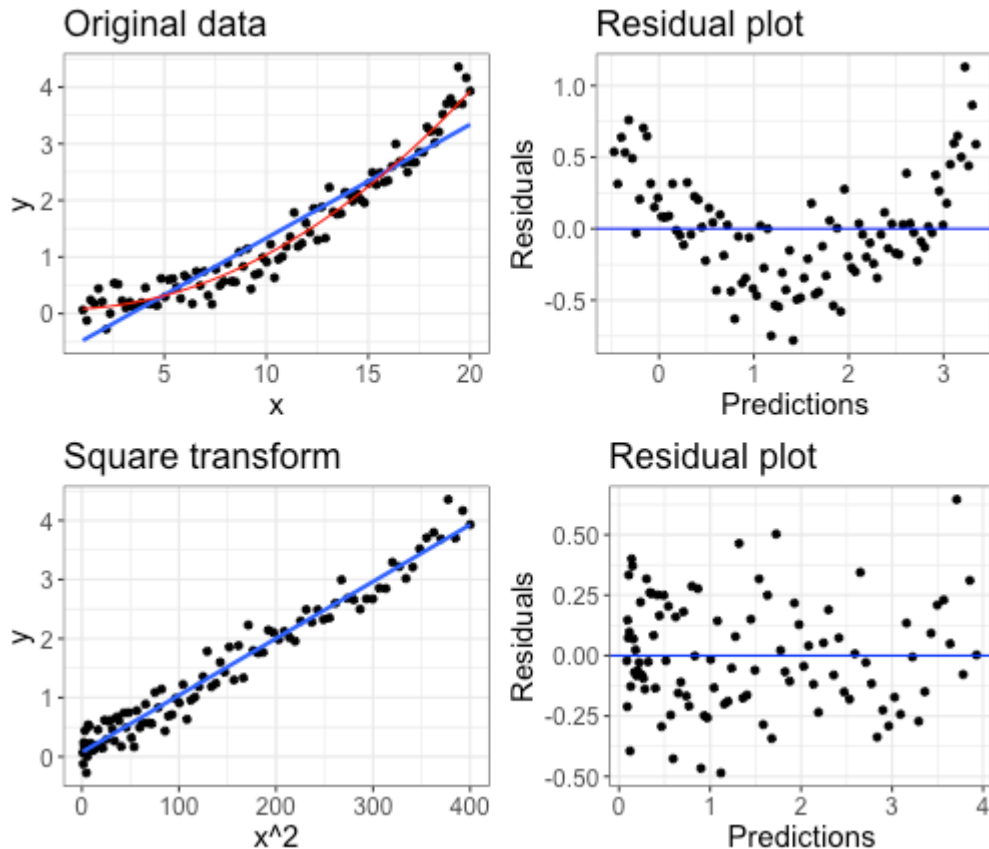
Example



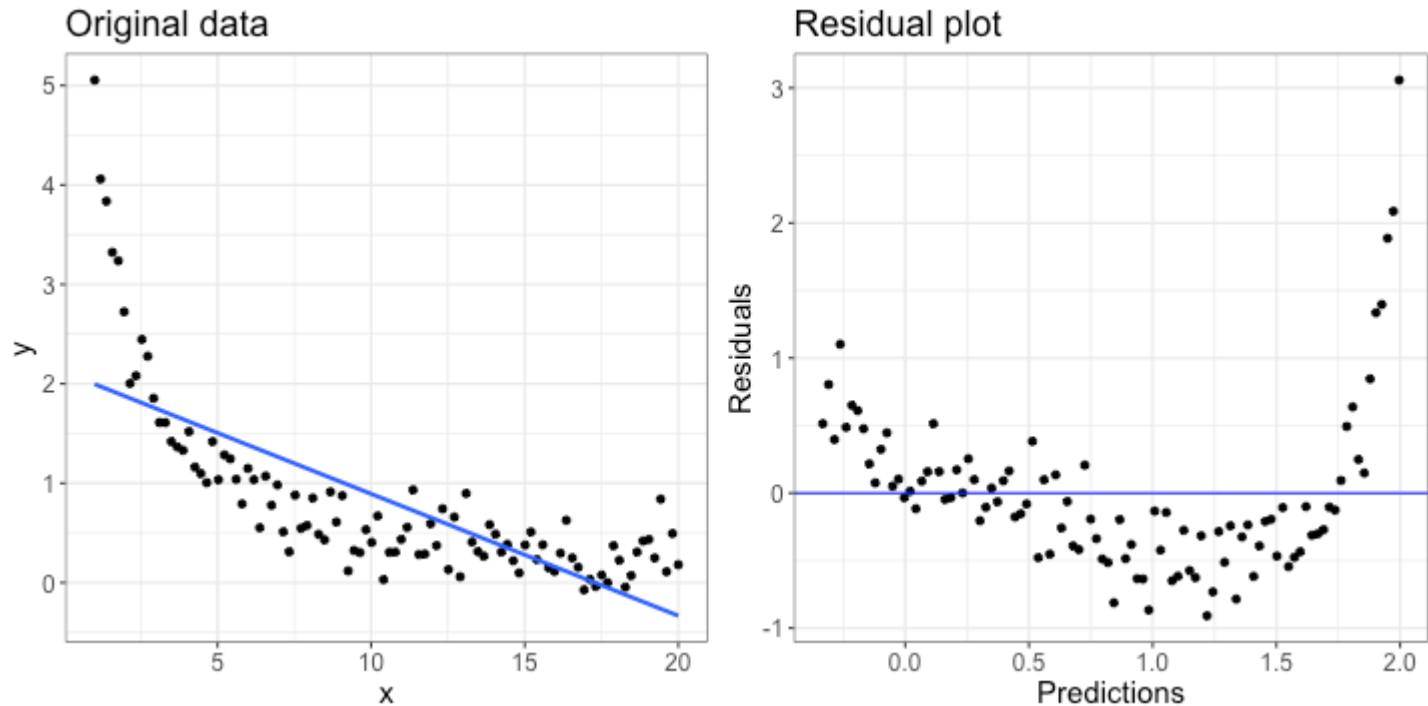
The relationship between x and y does not appear linear. How could we model that relationship instead?

Example: squaring X

$$Y = \beta_0 + \beta_1 X^2 + \varepsilon$$



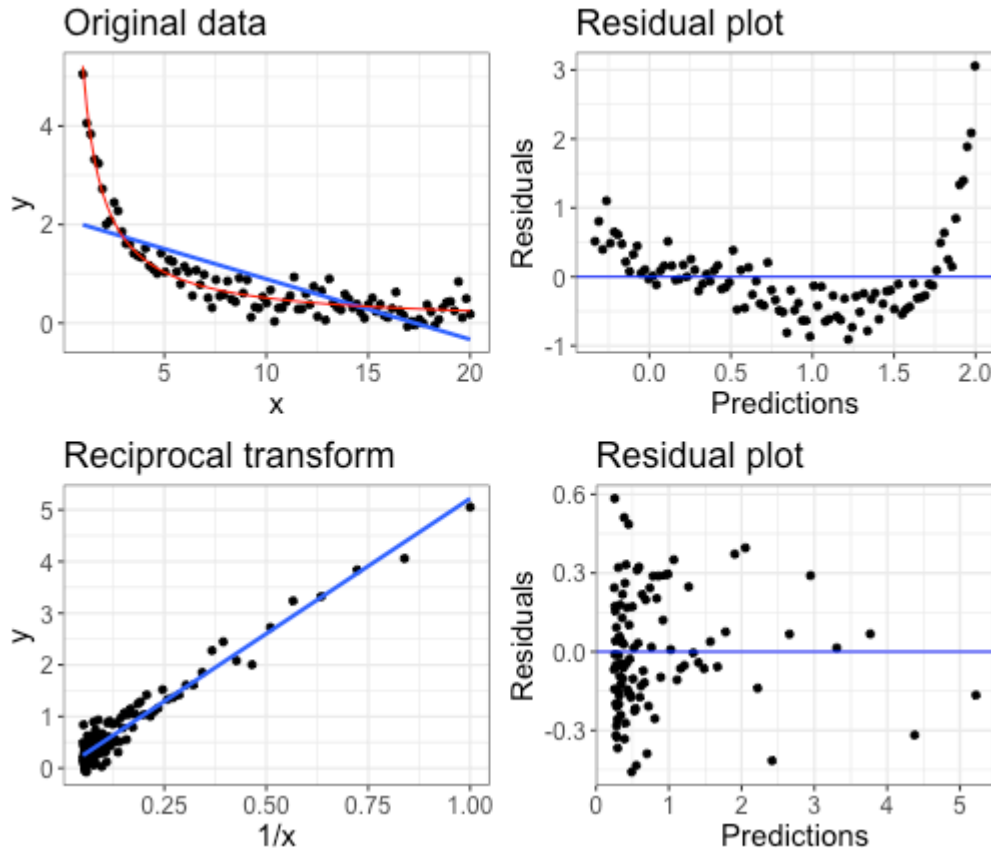
Example



The relationship between x and y does not appear linear. How could we model that relationship instead?

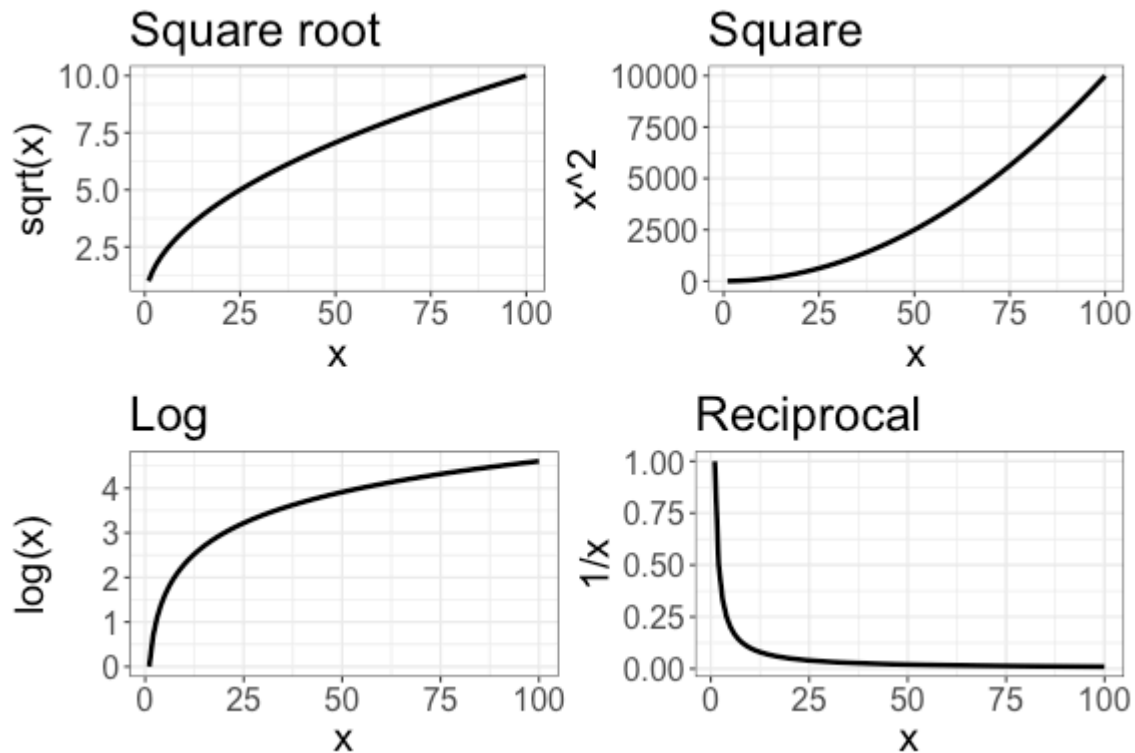
Example: reciprocal transform

$$Y = \beta_0 + \beta_1 (1/X) + \varepsilon$$

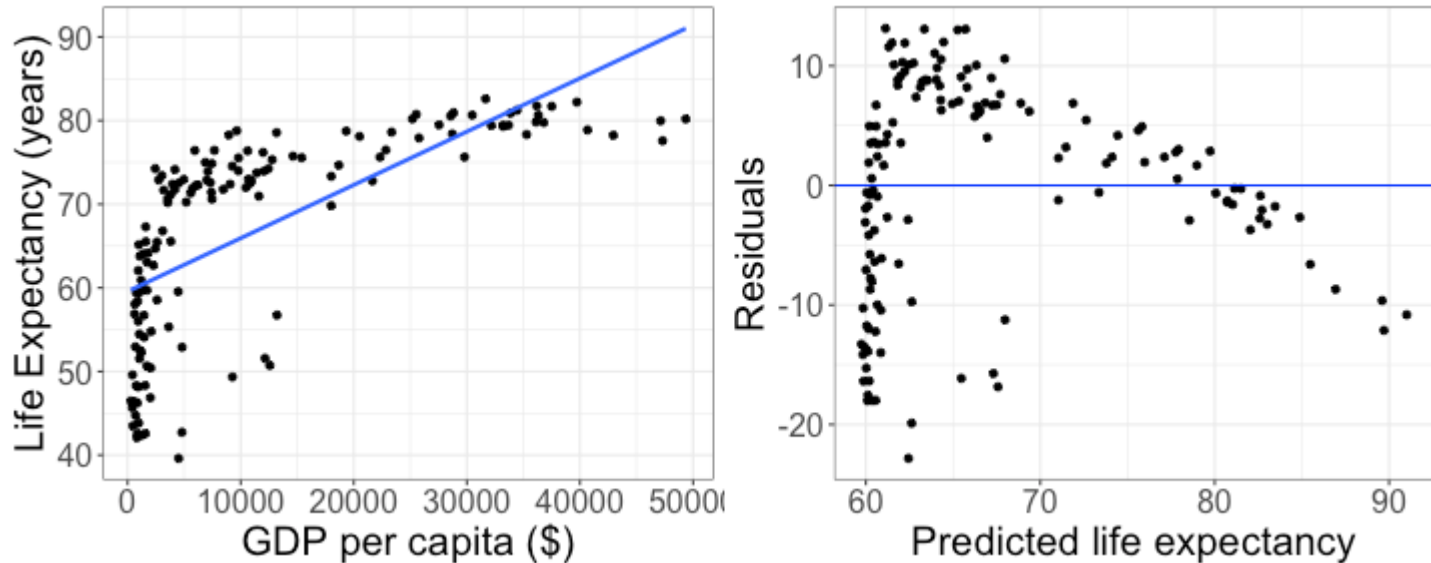


Choosing transformations

To model nonlinear relationships, try transformations for x that looks like the relationship

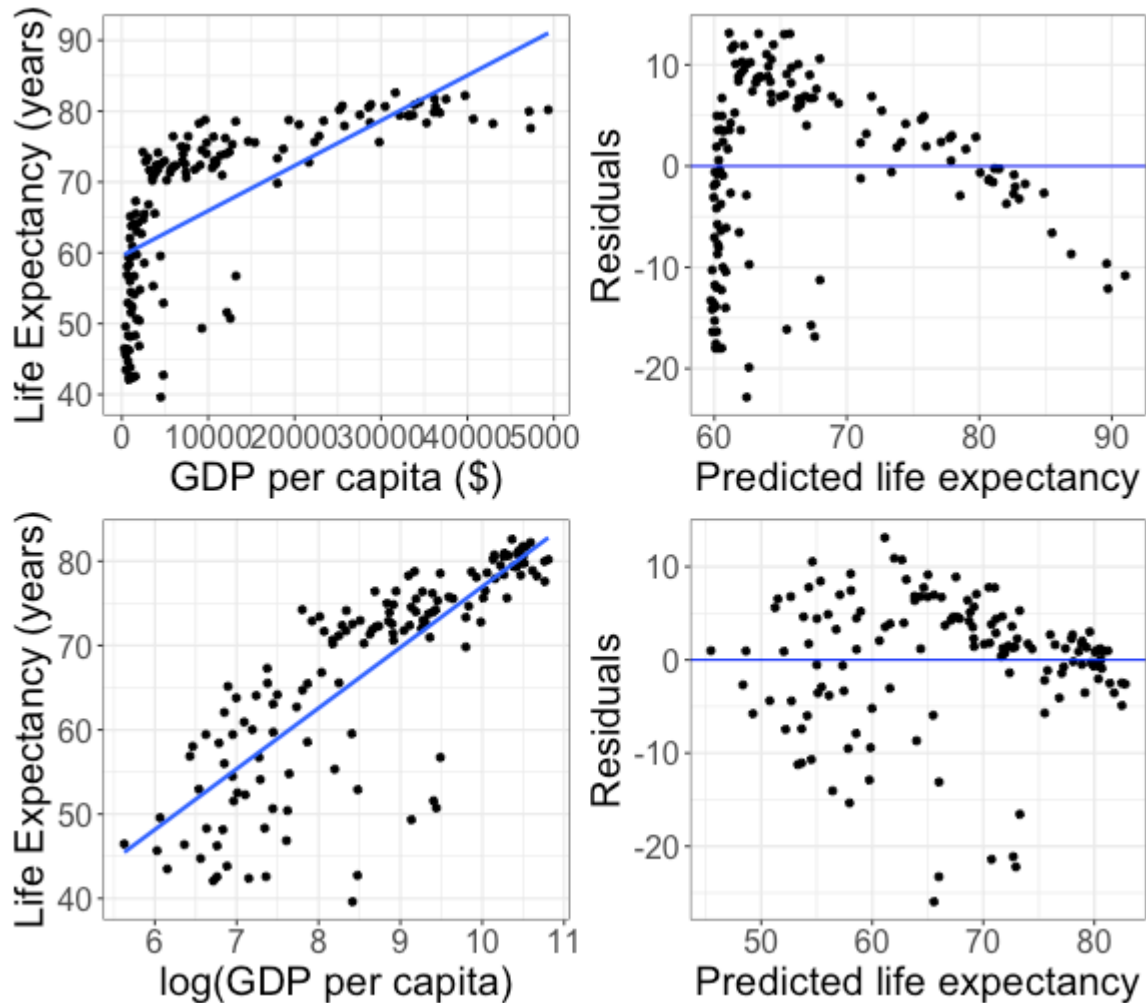


Example

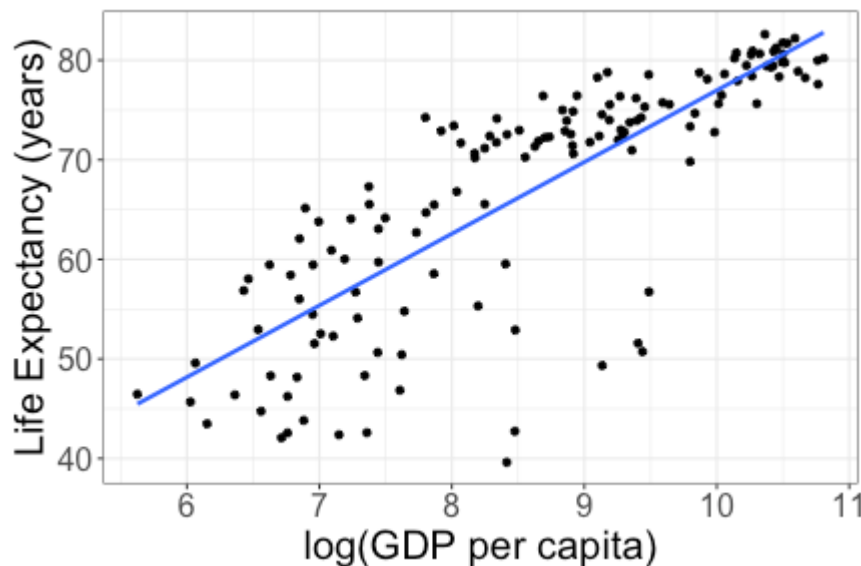


What transformation could I try for this data?

Log transformation



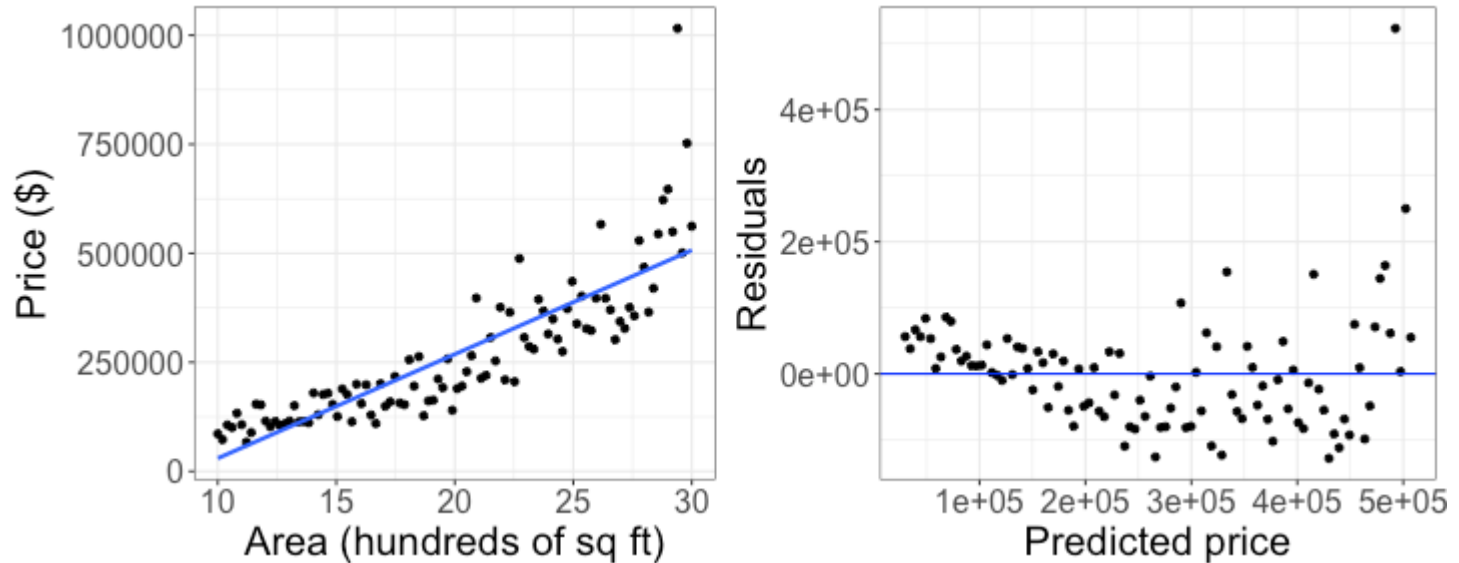
Log transformation



$$\widehat{\text{Life Expectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

What life expectancy would I predict for a GDP per capita of \$20000?

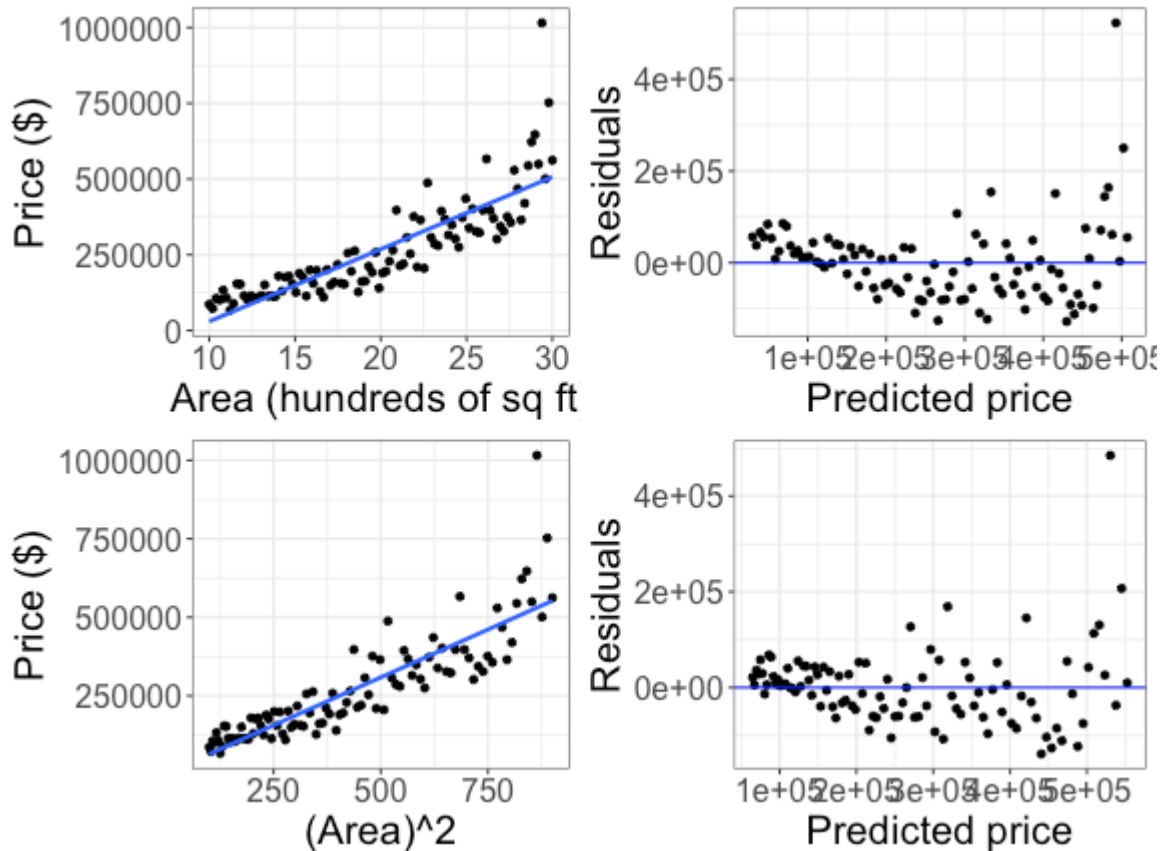
Example



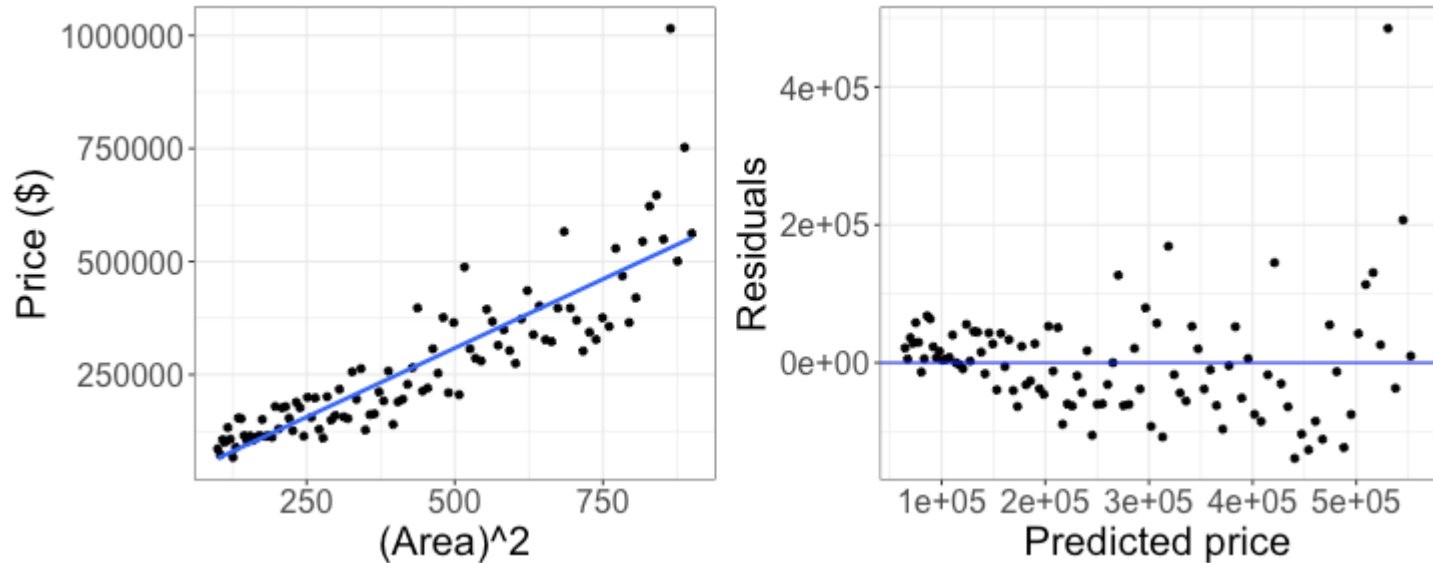
What transformation could I try for this data?

Trying squaring X

$$\text{Model: } Price = \beta_0 + \beta_1 (Area)^2 + \varepsilon$$



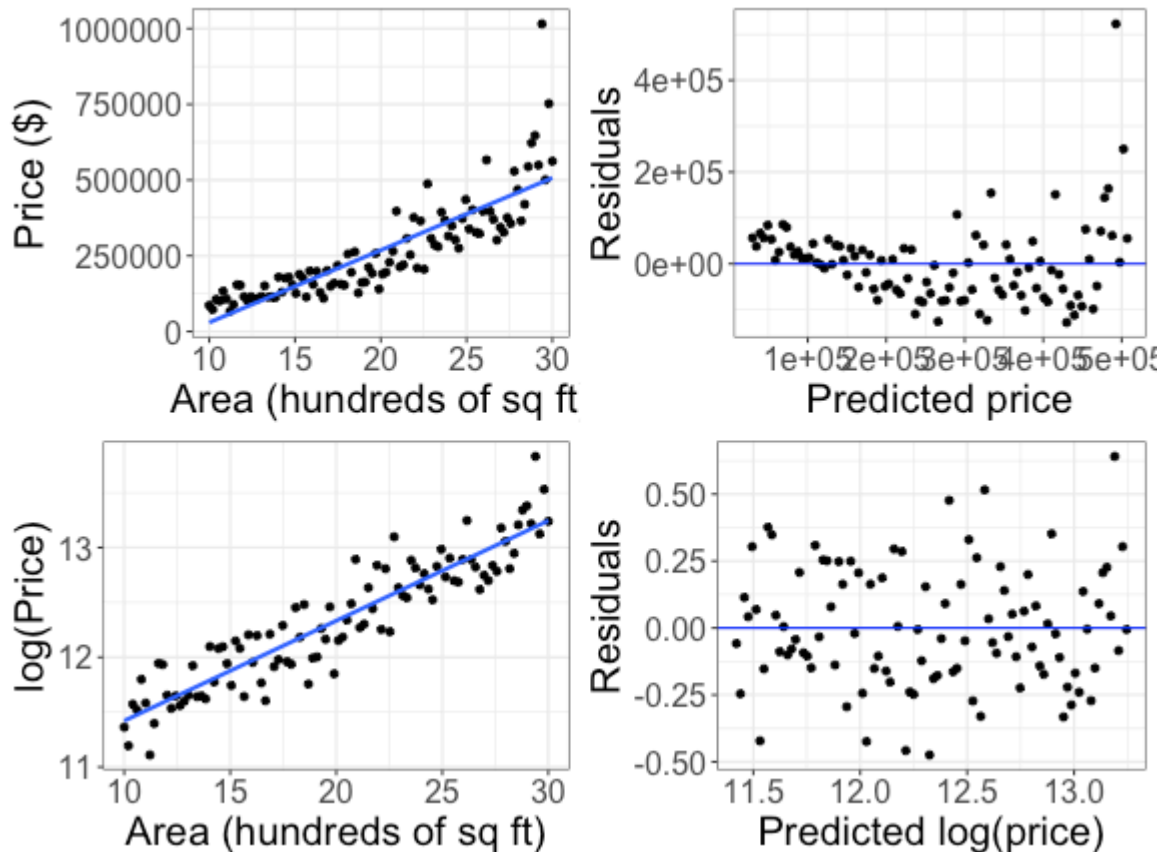
Trying squaring X



What do you notice about the residual plot after trying the transformation?

Another attempt: transforming the response!

$$\text{Model: } \log(\text{Price}) = \beta_0 + \beta_1 \text{Area} + \varepsilon$$



Log transformed response

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

+ Here Price is in \$, Area is in hundreds of square feet

How would I calculate the predicted *price* (in \$) for a house of 2000 square feet?

Class activity

Work on the handout, then we will discuss as a class

Class activity

$$\widehat{\text{Life Expectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

What is the predicted price for GDP per capita of \$20200?

Class activity

$$\widehat{\text{LifeExpectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

+ When GDP per capita = 20000: $\widehat{\text{LifeExpectancy}} = 76.255$

+ When GDP per capita = 20200: $\widehat{\text{LifeExpectancy}} = 76.327$

Change:

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

+ Price in \$, Area in hundres of sq ft

What is the predicted price for a house of 2100 square feet?

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

- + Price in \$, Area in hundres of sq ft
- + When Area = 20: $\widehat{\text{Price}} = 226386$
- + When Area = 21: $\widehat{\text{Price}} = 247954$

Change:

Class activity

$$\log(\widehat{\text{Price}}) = 10.51 + 0.091 \text{ Area}$$

- + Price in \$, Area in hundres of sq ft
- + When Area = 20: $\widehat{\text{Price}} = 226386$
- + When Area = 21: $\widehat{\text{Price}} = 247954$

$$\text{Change: } \frac{247954}{226386} = 1.095 = e^{0.091}$$